Assignment 18.4

Htno:2303A51291

Btno:05

Task 1: Movie Database API Integration

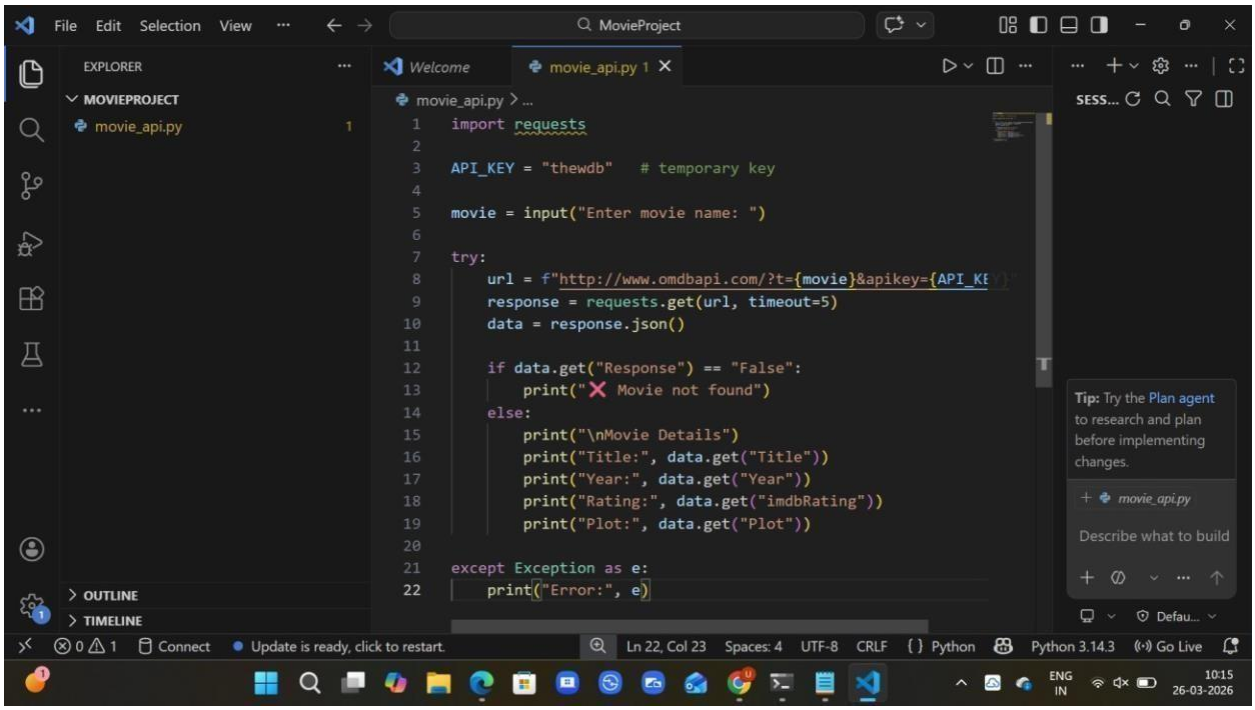
Prompt:Generate a Python script to fetch movie details using OMDB API.

The program should accept movie title as input and display title, year,

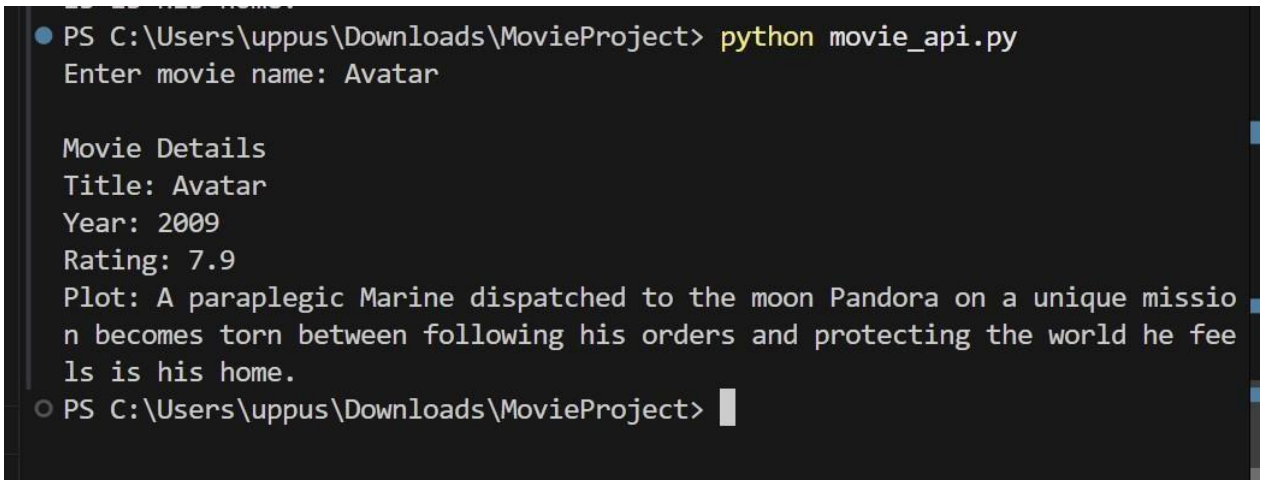
rating, and plot. Include proper error handling for invalid movie names,

API errors, and network issues.

Code:



Output:



Explanation:

-->The program uses requests library to connect to OMDB API.

-->It takes movie name as user input.

-->The API response is converted into JSON format.

-->Required details like title, year, rating, and plot are displayed.

-->Error handling is implemented for invalid movie names and network issues.

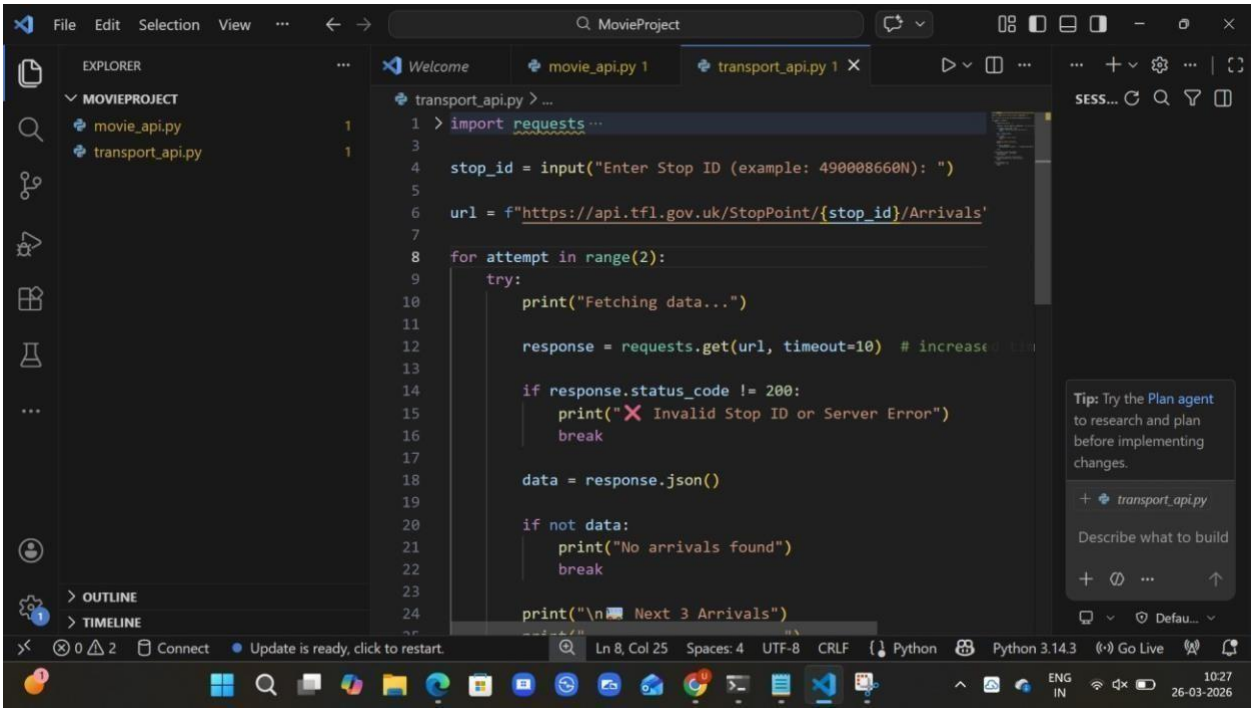
Task 2: Public Transport Schedule API Integration

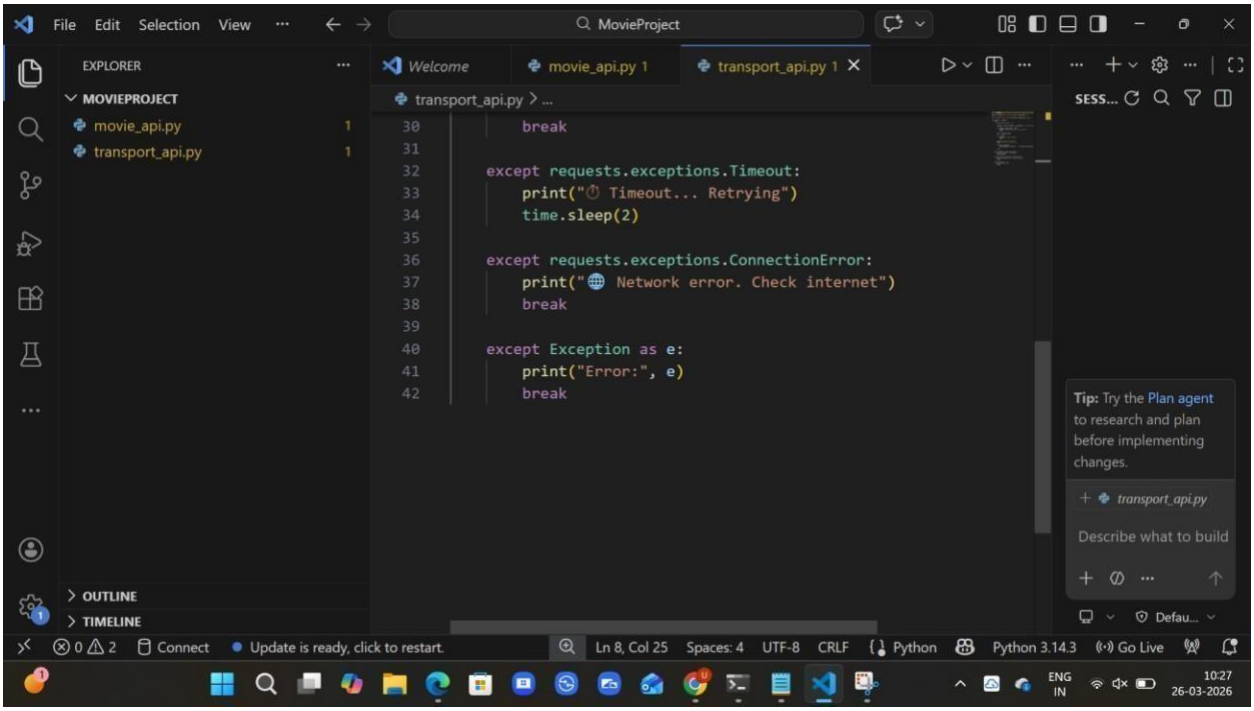
Prompt: Generate a Python script to fetch public transport arrival timings using an API.
The program should accept a stop ID as input and display the next 3 arrivals.

Include error handling for invalid stop ID, server issues, and timeout errors.

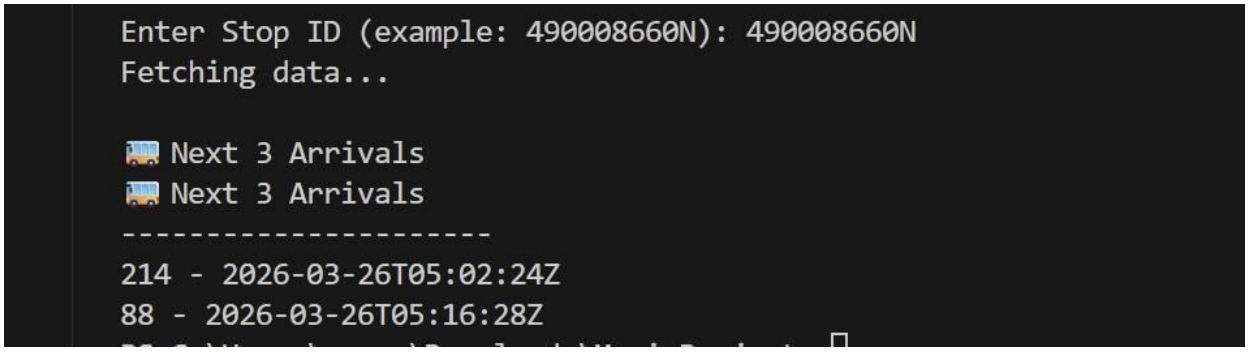
Also implement retry logic for failed API calls.

Code:


File Edit Selection View
MovieProject
EXPLORER
MOVIEPROJECT
movie_api.py 1
transport_api.py 1
Welcome
movie_api.py 1
transport_api.py 1
transport_api.py > ...
1 import requests ···
4 stop_id = input("Enter Stop ID (example: 490008660N): ")
6 url = f"https://api.tfl.gov.uk/StopPoint/{stop_id}/Arrivals"
8 for attempt in range(2):
9 try:
10 print("Fetching data...")
12 response = requests.get(url, timeout=10) # increase
14 if response.status_code != 200:
15 print("❌ Invalid Stop ID or Server Error")
16 break
18 data = response.json()
20 if not data:
21 print("No arrivals found")
22 break
24 print("\n Next 3 Arrivals")
Tip: Try the Plan agent to research and plan before implementing changes.
transport_api.py
Describe what to build
OUTLINE
TIMELINE
0 2 Connect Update is ready, click to restart.
Ln 8, Col 25 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN 10:27 26-03-2026


File Edit Selection View
MovieProject
EXPLORER
MOVIEPROJECT
movie_api.py 1
transport_api.py 1
Welcome
movie_api.py 1
transport_api.py 1
transport_api.py > ...
30 break
32 except requests.exceptions.Timeout:
33 print("⏱ Timeout... Retrying")
34 time.sleep(2)
36 except requests.exceptions.ConnectionError:
37 print("🌐 Network error. Check internet")
38 break
40 except Exception as e:
41 print("Error:", e)
42 break
Tip: Try the Plan agent to research and plan before implementing changes.
transport_api.py
Describe what to build
OUTLINE
TIMELINE
0 2 Connect Update is ready, click to restart.
Ln 8, Col 25 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN 10:27 26-03-2026

Output:



Explanation:

-->The program uses the requests library to connect to a transport API and fetch arrival data.

-->It takes stop ID as user input to retrieve specific station details.

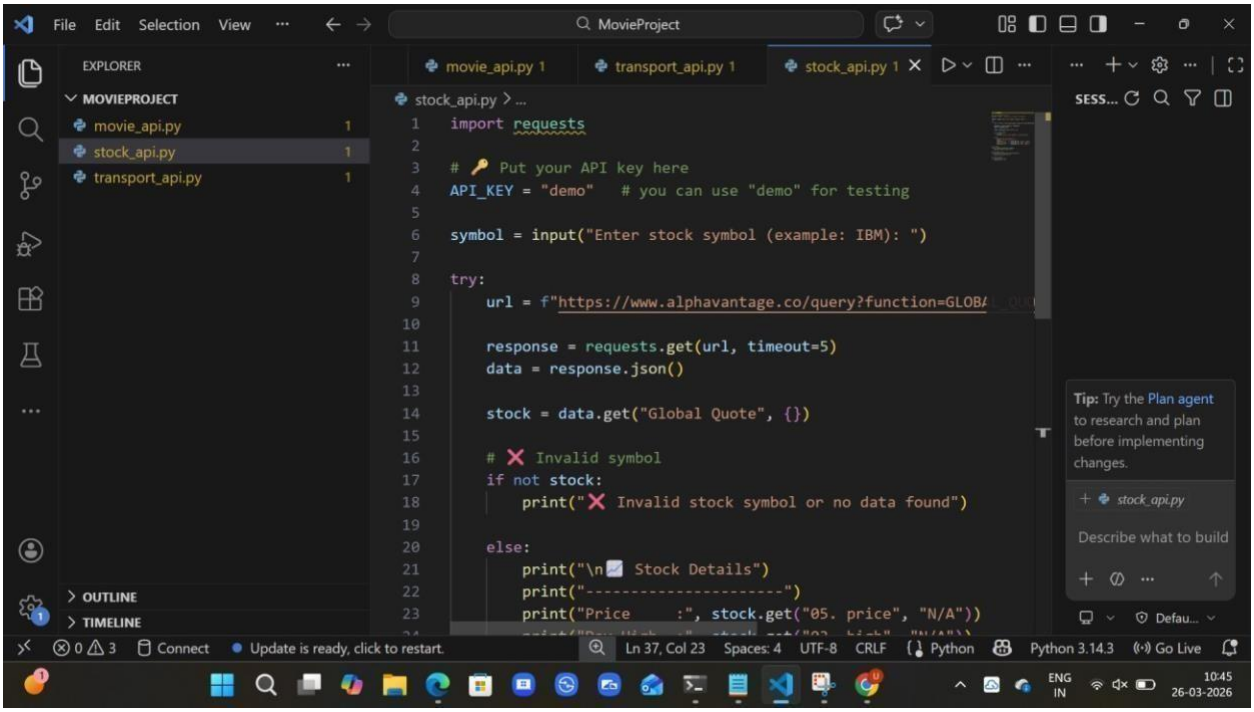
-->The API response is converted into JSON format and processed.

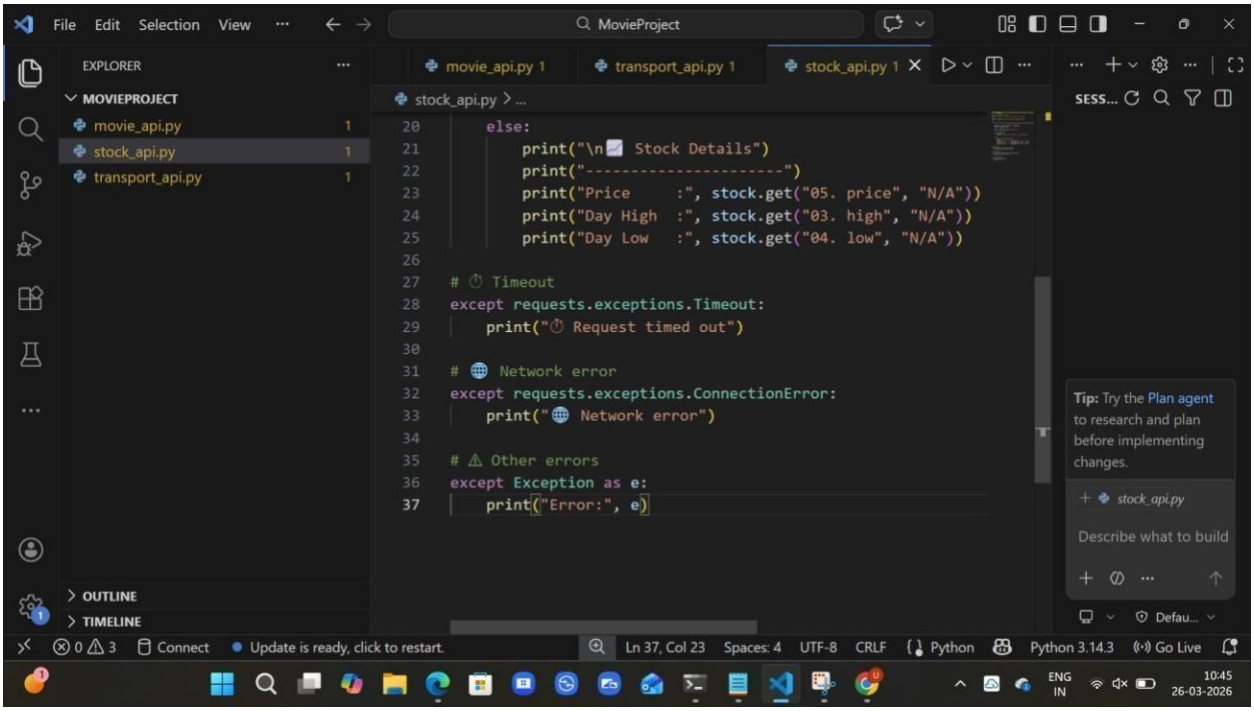
-->It displays the next 3 arrival timings in a readable format.

Task 3: Stock Market Data API Integration

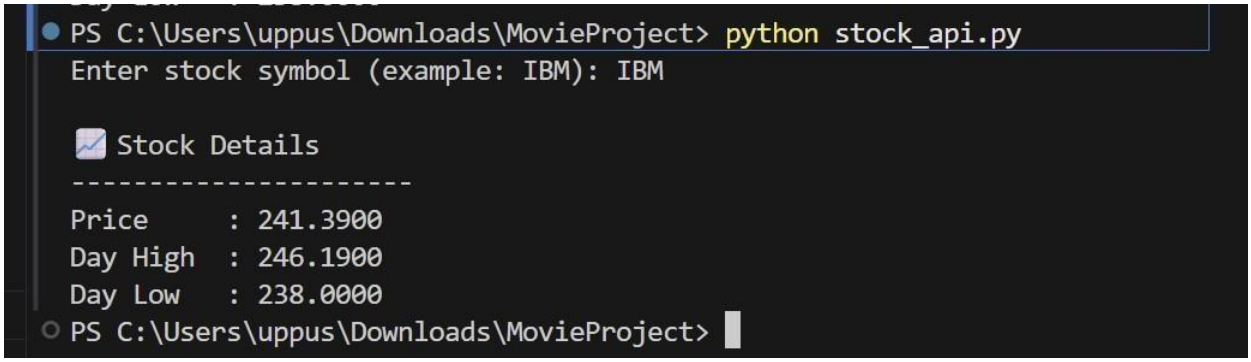
Prompt: Generate a Python script to fetch stock market data using a financial API. The program should accept a stock symbol as input and display current price, day high, and day low. Include error handling for invalid symbols, API rate limits, and missing data in the API response.

Code:


File Edit Selection View
MovieProject
EXPLORER
MOVIEPROJECT
movie_api.py 1
stock_api.py 1
transport_api.py 1
movie_api.py 1
transport_api.py 1
stock_api.py 1
stock_api.py > ...
import requests
# 🔑 Put your API key here
API_KEY = "demo" # you can use "demo" for testing
symbol = input("Enter stock symbol (example: IBM): ")
try:
url = f"https://www.alphavantage.co/query?function=GLOB
response = requests.get(url, timeout=5)
data = response.json()
stock = data.get("Global Quote", {})
# ❌ Invalid symbol
if not stock:
print("❌ Invalid stock symbol or no data found")
else:
print("\n📈 Stock Details")
print("----------------------")
print("Price :", stock.get("05. price", "N/A"))
Tip: Try the Plan agent to research and plan before implementing changes.
stock_api.py
Describe what to build
OUTLINE
TIMELINE
0 3 Connect Update is ready, click to restart.
Ln 37, Col 23 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN
10:45 26-03-2026


File Edit Selection View
MovieProject
EXPLORER
MOVIEPROJECT
movie_api.py 1
stock_api.py 1
transport_api.py 1
movie_api.py 1
transport_api.py 1
stock_api.py 1
stock_api.py > ...
else:
print("\n📈 Stock Details")
print("----------------------")
print("Price :", stock.get("05. price", "N/A"))
print("Day High :", stock.get("03. high", "N/A"))
print("Day Low :", stock.get("04. low", "N/A"))
# ⏱ Timeout
except requests.exceptions.Timeout:
print("⏱ Request timed out")
# 🌐 Network error
except requests.exceptions.ConnectionError:
print("🌐 Network error")
# ⚠ Other errors
except Exception as e:
print("Error:", e)
Tip: Try the Plan agent to research and plan before implementing changes.
stock_api.py
Describe what to build
OUTLINE
TIMELINE
0 3 Connect Update is ready, click to restart.
Ln 37, Col 23 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN
10:45 26-03-2026

Output:



Explaantion:

-->The program uses the requests library to fetch stock data from a financial API.

-->It takes stock symbol as input from the user (e.g., IBM).

-->The API response is converted into JSON format and processed safely.

-->It displays important details like current price, day high, and day low.

-->Error handling is implemented for invalid symbols, network issues, and missing JSON fields.

Task 4: **Geolocation API (IP Address Lookup)**

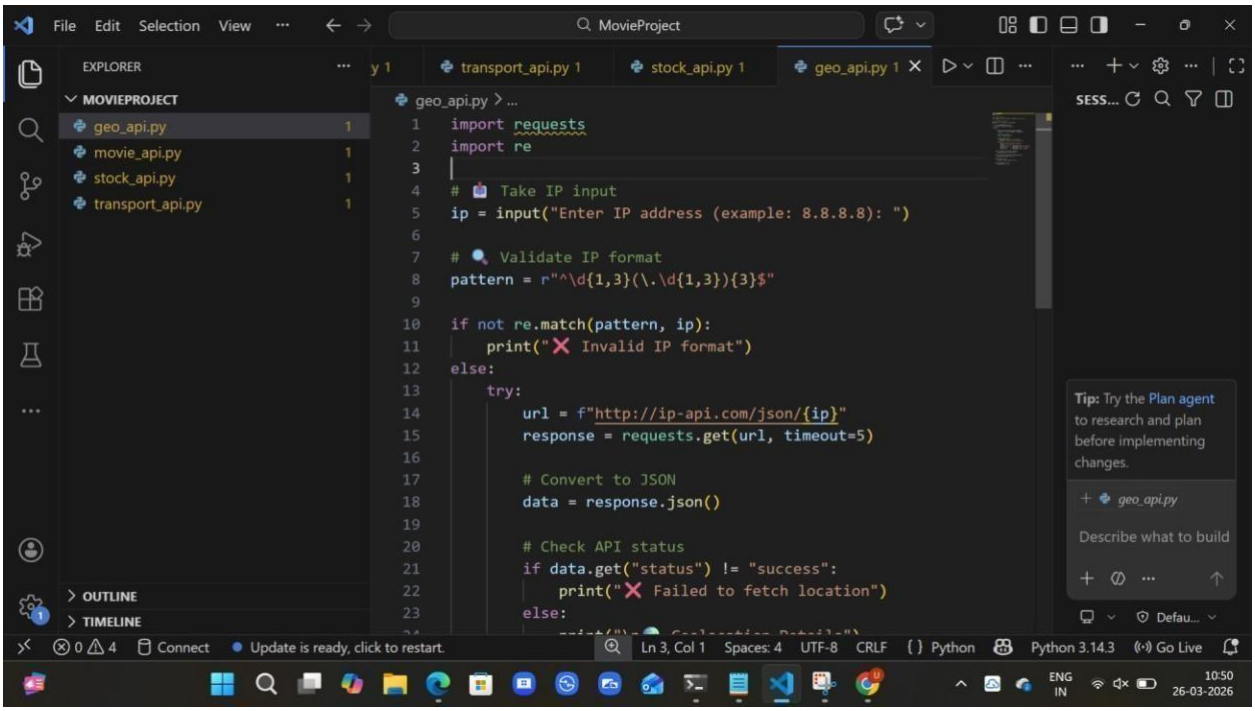
Prompt: Generate a Python script to fetch geolocation details using an IP address.

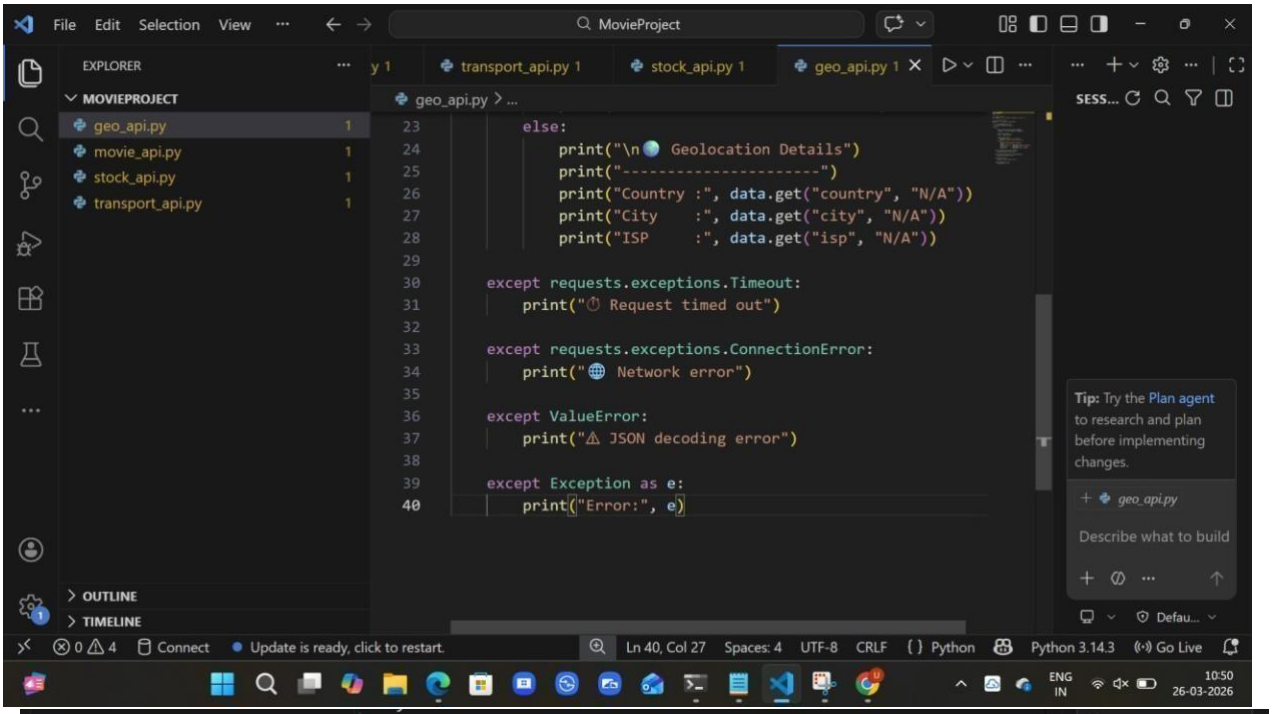
The program should accept an IP address as input and display country, city, and ISP.

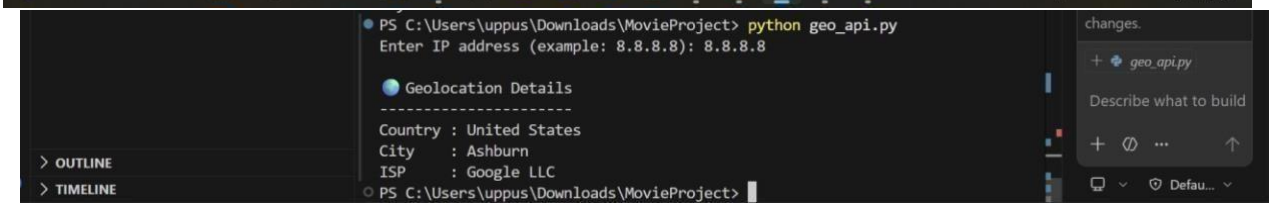
Include validation for IP format and error handling for invalid input, API failure,

and JSON decoding errors.

Code:







Explanation:

-->The program accepts an IP address as user input.

Output:

-->It validates the IP format using regular expressions (regex).

-->It uses the requests library to connect to a geolocation API.

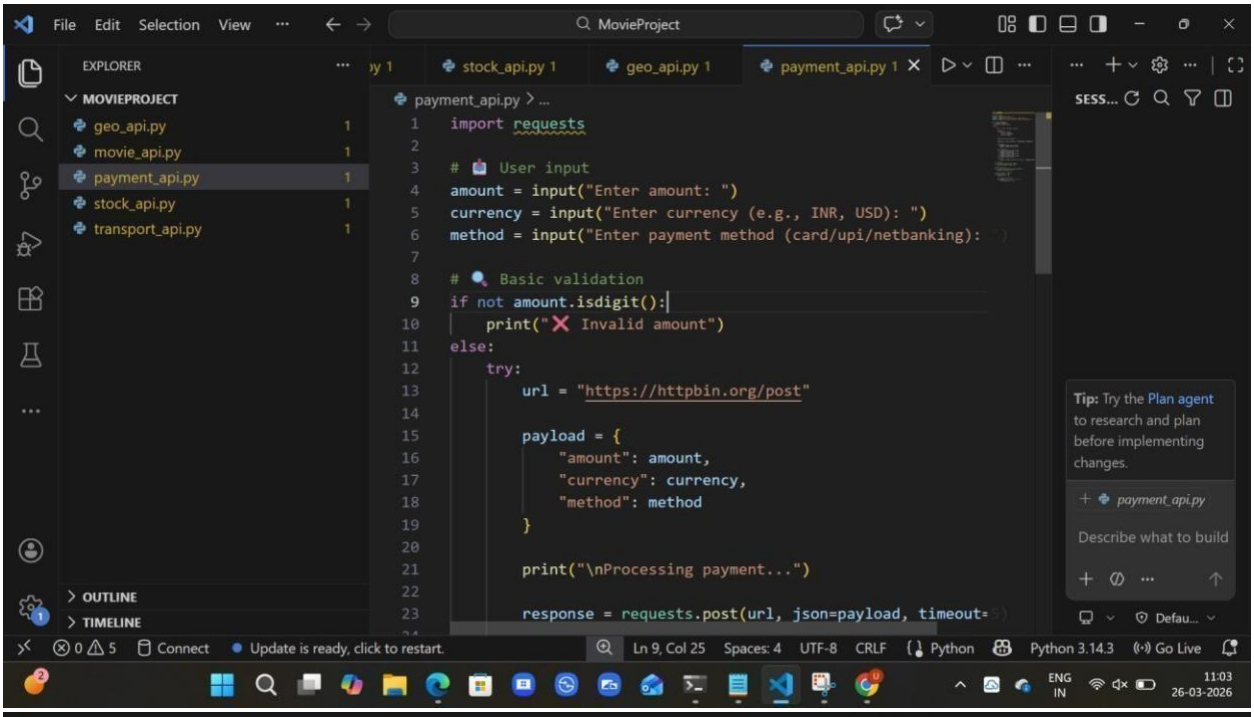
-->The API response is converted into JSON format and parsed.

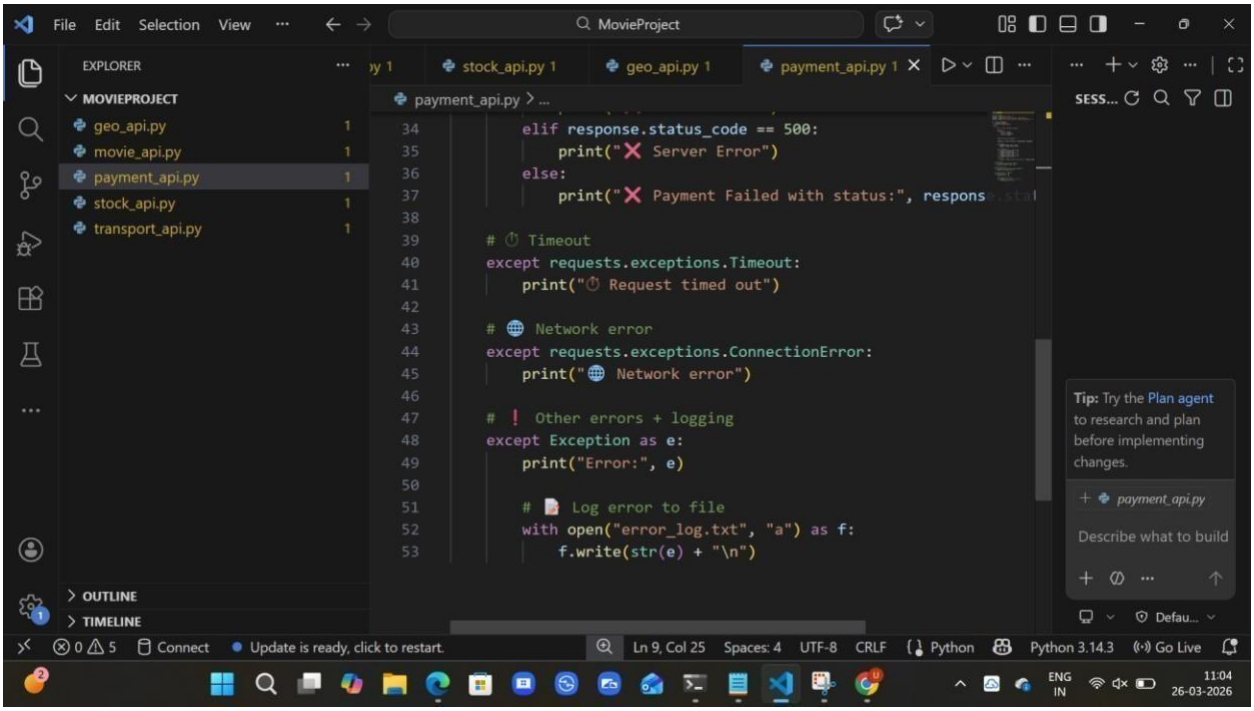
-->Error handling is implemented for invalid IP, network issues, API failure, and JSON errors.

Task 5: Real-Time Application – Payment Gateway Simulation API

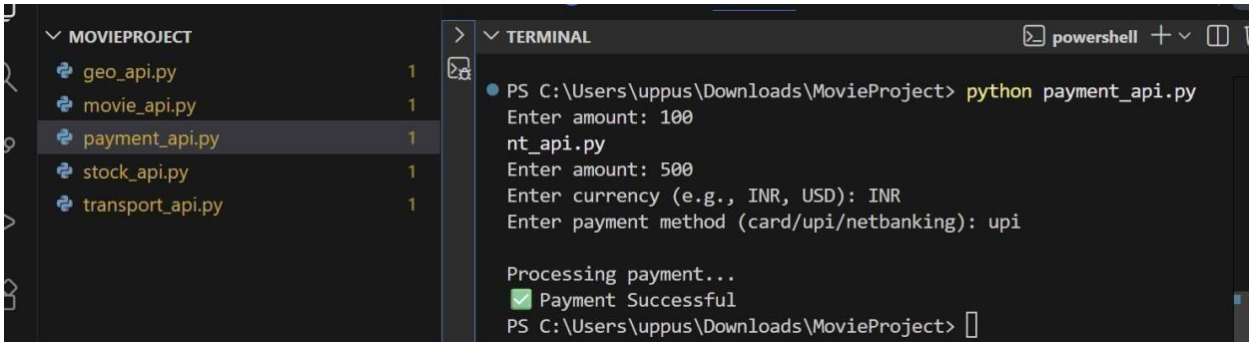
Prompt: Generate a Python script to simulate payment processing using a sandbox API. The program should send a POST request with amount, currency, and payment method. Include input validation and handle errors such as invalid payment details, HTTP status errors (400, 401, 500), and network issues. Also log failed transactions into a file.

Code:


File Edit Selection View
MovieProject
EXPLORER
MOVIEPROJECT
geo_api.py 1
movie_api.py 1
payment_api.py 1
stock_api.py 1
transport_api.py 1
stock_api.py 1
geo_api.py 1
payment_api.py 1
payment_api.py > ...
import requests
# 👤 User input
amount = input("Enter amount: ")
currency = input("Enter currency (e.g., INR, USD): ")
method = input("Enter payment method (card/upi/netbanking): ")
# 🔍 Basic validation
if not amount.isdigit():
print("❌ Invalid amount")
else:
try:
url = "https://httpbin.org/post"
payload = {
"amount": amount,
"currency": currency,
"method": method
}
print("\nProcessing payment...")
response = requests.post(url, json=payload, timeout=
Tip: Try the Plan agent to research and plan before implementing changes.
payment_api.py
Describe what to build
OUTLINE
TIMELINE
0 5 Connect Update is ready, click to restart.
Ln 9, Col 25 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN 11:03 26-03-2026


File Edit Selection View
MovieProject
EXPLORER
MOVIEPROJECT
geo_api.py 1
movie_api.py 1
payment_api.py 1
stock_api.py 1
transport_api.py 1
payment_api.py > ...
elif response.status_code == 500:
print("❌ Server Error")
else:
print("❌ Payment Failed with status:", respons
⏱ Timeout
except requests.exceptions.Timeout:
print("⏱ Request timed out")
# 🌐 Network error
except requests.exceptions.ConnectionError:
print("🌐 Network error")
# ! Other errors + logging
except Exception as e:
print("Error:", e)
# 📝 Log error to file
with open("error_log.txt", "a") as f:
f.write(str(e) + "\n")
Tip: Try the Plan agent to research and plan before implementing changes.
payment_api.py
Describe what to build
OUTLINE
TIMELINE
0 5 Connect Update is ready, click to restart.
Ln 9, Col 25 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN 11:04 26-03-2026

Output:



Explanation:

-->The program collects payment details (amount, currency, method) from the user.

-->It sends a POST request to a sandbox API to simulate payment processing.

-->Input validation is performed to ensure the amount is valid before processing.

-->It handles errors such as HTTP status errors (400, 401, 500), timeout, and network issues.

-->Failed transactions are logged into a file for future reference and debugging.